Documentation

Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes mutant

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Cite
Wamp, S., Holland, G., Rismondo, J., & Halbedel, S. (2026). Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes mutant [Data set]. Zenodo.
https://doi.org/10.5281/zenodo.18349471

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Abstract
The dataset "Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes mutant" provides the raw images and data for the manuscript "On the role of cell chaining in the attenuation of a Listeria monocytogenes divIVA mutant" (https://doi.org/10.64898/2026.01.05.697648).

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Information on the dataset and context of origin

The dataset “Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes mutant” provides the raw images and data, which form the basis for the figures of the manuscript “On the role of cell chaining in the attenuation of a Listeria monocytogenes divIVA mutant”. In-depth information on the background of the work, on methodology and on the analysis of the results can be found in

> Wamp, S., Holland, G., Rismondo. J., and Halbedel, S. (2026). On the role of cell chaining in the attenuation of a Listeria monocytogenes divIVA mutant. bioRxiv.
> https://doi.org/10.64898/2026.01.05.697648

Administrative and organizational information

The publicaton “Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes mutant”, is a joined work of the Consultant Laboratory for Listeria in Unit 11| Enteropathogenic Bacteria and Legionella of the Robert Koch Institute, the Unit ZBS 4| Advanced Light and Electron Microscopy and the Institute for Medical Microbiology and Hospital Hygiene at the Otto von Guericke University.
Questions regarding the content of the data can be addressed directly to the corresponding author Sven Halbedel (halbedels@rki.de).
The publication of the data as well as the quality management of the (meta-)data is done by the unit MF 4 | Domain Specific Data and Research Data Management. Questions regarding data management and the publication infrastructure can be directed to the Open Data Team of the unit MF4 at OpenData@rki.de.

Overview over the *L. monocytogenes* strains used for the manuscript

The following strains were used for the manuscript. Detailed methodology on their generation can be found in the methods section of the manuscript (https://doi.org/10.64898/2026.01.05.697648).

Strain name	Genetic background	Description
EGD-e	wt	Wildtype strain *L. monocytogenes*, used as control
LMSW49	*imreB*	MreB depletion strain, mreB expression can be induced upon IPTG-treatment
LMJR183	*iezrA*	EzrA depletion strain, ezrA expression can be induced upon IPTG-treatment
LMS2	*ΔdivIVA*	divIVA deletion strain
LMSW207	*ΔdivIVA imreB*	divIVA deletion and MreB depletion strain, mreB expression can be induced upon IPTG-treatment

Strain name	Genetic background	Description
LMSW208	Δ*divIVA iezrA*	divIVA deletion and EzrA depletion strain, ezrA expression can be induced upon IPTG-treatment
BUG2214	Δ*prfA*	prfA deletion strain
LMS30	*idivIVA*	DivIVA depletion strain, divIVA expression can be induced upon IPTG-treatment
LMJD20	wt	DsRed producing *L. monocytogenes* strain, used as control
LMSF2	Δ*actA*	DsRed producing, actA deletion strain
LMSW205	Δ*divIVA*	DsRed producing, divIVA deletion strain
LMSW202	*iezrA*	DsRed producing, EzrA depletion strain, ezrA expression can be induced upon IPTG-treatment
LMSW203	*imreB*	DsRed producing, MreB depletion strain, mreB expression can be induced upon IPTG-treatment
BUG3057	*prfA**	*L. monocytogenes* strain that encodes a PrfA variant that is constitutively active causing constitutive overexpression of PrfA-dependent virulence genes
LMSW251	Δ*divIVA prfA**	divIVA deletion strain that encodes a PrfA variant that is constitutively active causing constitutive overexpression of PrfA-dependent virulence genes
LMSW222	Δ*divIVA secA2 M95I*	divIVA deletion strain with secA2 mutation (M95I)
LMSW223	Δ*divIVA secA2 T123A*	divIVA deletion strain with secA2 mutation (T123A)

Content and structure of the raw supplemented raw images and data

The raw data and relevant calculations that form the basis for the manuscript's figures are provided as .xlsx files.

The raw microscopy images, as well as uncropped images from a western blot, a gel electrophoresis and culture plates are provided in the "Images" subfolder as .tif files.
> Images/

The naming and content of the image and data files are explained in the tables below per figure.

Figure 1

Figure	Description	File
Figure 1A	Growth curves of the strains EGD-e (wt) and LMSW49 (*imreB*) ± IPTG	Figure_1A_raw_data.xlsx
Figure 1B	Uncropped Western blot showing MreB and DivIVA expression levels in the strains EGD-e (wt) and LMSW49 (*imreB*) ± IPTG	Images/Figure_1B_uncropped_blot_images.tif
Figure 1C	Fluorescence microscopy image of EGD-e (wt, nile red)	Images/Figure_1C_wt_raw_image.tif
Figure 1C	Fluorescence microscopy image of LMSW49 (*imreB*) with IPTG (nile red)	Images/Figure_1C_imreB+IPTG_raw_image.tif
Figure 1C	Fluorescence microscopy image of LMSW49 (*imreB*) without IPTG (nile red)	Images/Figure_1C_imreB-IPTG_raw_image.tif

Figure 2

Figure	Description	File
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Figure 2B	Fluorescence microscopy image of EGD-e (wt, nile red)	Images/Figure_2B_wt_nile_red_raw_image.tif
Figure 2B	Microscopy image of EGD-e (wt) in phase contrast	Images/Figure_2B_wt_pc_raw_image.tif

Figure	Description	File
Figure 2B	Fluorescence microscopy image of i*ezrA* without IPTG (nile red)	Images/Figure_2B_iezrA-IPTG_nile_red_raw_image.tif
Figure 2B	Microscopy image of i*ezrA* without IPTG in phase contrast	Images/Figure_2B_iezrA-IPTG_pc_raw_image.tif
Figure 2B	Fluorescence microscopy image of i*ezrA* with IPTG (nile red)	Images/Figure_2B_iezrA+IPTG_nile_red_raw_image.tif
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Figure 3

Figure	Description	File
Figure 3A	Growth curves of the strains EGD-e (wt) and LMS2 (Δ*divIVA*)	Figure_3A_raw_data.xlsx
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Figure 4

Figure	Description	File
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Figure 4	Scanning electron microscopy image of LMS2 (Δ*divIVA*)	Images/Figure_4_divIVA_raw_image.tif
Figure 4	Scanning electron microscopy image of LMSW49 (i*mreB*) without IPTG	Images/Figure_4_mreB_image.tif

Figure	Description	File
Figure 4	Scanning electron microscopy image of LMJR183 (i*ezrA*) without IPTG	Images/Figure_4_ezrA_image.tif

Figure 5

Figure	Description	File
Figure 5A	Fluorescence microscopy image of Δ*divIVA* i*mreB* (LMSW207, nile red) without IPTG	Images/Figure_5A_divIVA_mreB_raw_image.tif
Figure 5A	Fluorescence microscopy image of Δ*divIVA* i*ezrA* (LMSW208, nile red) without IPTG	Images/Figure_5A_divIVA_ezrA_raw_image.tif
Figure 5B	Scanning electron microscopy image of LMSW207 (Δ*divIVA* i*mreB*) without IPTG	Images/Figure_5B_divIVA_mreB_raw_image.tif
Figure 5B	Scanning electron microscopy image of LMSW208 (Δ*divIVA* i*ezrA*) without IPTG	Images/Figure_5B_divIVA_ezrA_raw_image.tif

Figure 6

Figure	Description	File
Figure 6A	(Relative) invasion rates into hepatocytes by EGD-e (wt), BUG2214 (Δ*prfA*), LMS2 (Δ*divIVA*), LMSW49 (i*mreB*) and LMJR183 (i*ezrA*)	Figure_6A_raw_data.xlsx
Figure 6B	Intracellular growth in macrophages of EGD-e (wt), BUG2214 (Δ*prfA*), LMS2 (Δ*divIVA*), LMSW49 (i*mreB*) and LMJR183 (i*ezrA*)	Figure_6B_raw_data.xlsx
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Figure 6D	Fluorence microscopy image of macrophages infected with	Images/Figure_6D_wt_raw_image.tif

Figure	Description	File
	LMJD20 (wt)	
Figure 6D	Fluorence microscopy image of macrophages infected with LMSF2 (Δ*actA*)	Images/Figure_6D_actA_raw_image.tif
Figure 6D	Fluorence microscopy image of macrophages infected with LMSW205 (Δ*divIVA*)	Images/Figure_6D_divIVA_raw_image.tif
Figure 6D	Fluorence microscopy image of macrophages infected with LMSW202 (i*ezrA*) without IPTG	Images/Figure_6D_ezrA_raw_image.tif
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Figure 7

Figure	Description	File
Figure 7A	(Relative) invasion rates into hepatocytes by EGD-e (wt), BUG2214 (Δ*prfA*), BUG3057 (*prfA**), LMS2 (Δ*divIVA*) and LMSW251 (*ΔdivIVA prfA**)	Figure_7A_raw_data.xlsx
Figure 7B	Intracellular growth in macrophages of EGD-e (wt), BUG2214 (Δ*prfA*), BUG3057 (*prfA**), LMS2 (Δ*divIVA*) and LMSW251 (Δ*divIVA prfA**)	Figure_7B_raw_data.xlsx
Figure 7C	Plate photographies of plaque formation assays with EGD-e (wt), BUG2214 (Δ*prfA*), BUG3057 (*prfA**), LMS2 (Δ*divIVA*) and LMSW251 (Δ*divIVA prfA**)	Images/Figure_7C_raw_image.tif

Figure 8

Figure	Description	File
Figure 8B	Fluorescence microscopy image of EGD-e (wt, nile red)	Images/Figure_8B_wt_raw_image.tif

Figure	Description	File
Figure 8B	Fluorescence microscopy image of LMS2 (Δ*divIVA*, nile red)	Images/Figure_8B_divIVA_raw_image.tif
Figure 8B	Fluorescence microscopy image of LMSW222 (Δ*divIVA secA2 M95I*, nile red)	Images/Figure_8B_divIVA_secA2_M95I_raw_image.tif
Figure 8B	Fluorescence microscopy image of LMSW223 (Δ*divIVA secA2 T123A*, nile red)	Images/Figure_8B_divIVA_secA2_T123A_raw_image.tif
Figure 8C	SDS-PAGE image of EGD-e (wt), LMS2 (Δ*divIVA*), LMSW222 (Δ*divIVA secA2 M95I*) and LMSW223 (Δ*divIVA secA2 T123A*)	Images/Figure_8C_uncropped_gel_image.tif
Figure 8D	Images of swarming agar plates of EGD-e (wt), LMS2 (Δ*divIVA*), LMSW222 (Δ*divIVA secA2 M95I*) and LMSW223 (Δ*divIVA secA2 T123A*)	Images/Figure_8D_raw_image.tif
Figure 8E	Plate photographies of plaque formation assays with EGD-e (wt), LMS2 (Δ*divIVA*), LMSW222 (Δ*divIVA secA2 M95I*) and LMSW223 (Δ*divIVA secA2 T123A*)	Images/Figure_8E_raw_image.tif
Figure 8F	(Relative) invasion rates into hepatocytes by EGD-e (wt), LMS2 (Δ*divIVA*), LMSW222 (Δ*divIVA secA2 M95I*) and LMSW223 (Δ*divIVA secA2 T123A*)	Figure_8F_raw_data.xlsx
Figure 8G	Intracellular growth in macrophages of EGD-e (wt), LMS2 (Δ*divIVA*), LMSW222 (Δ*divIVA secA2 M95I*) and LMSW223 (Δ*divIVA secA2 T123A*)	Figure_8G_raw_data.xlsx

Supplementary Figure S1

Figure	Description	File
Figure S1	Scanning electron microscopy image of EGD-e (wt)	Images/Figure_S1_wt_raw_image.tif

Figure	Description	File
Figure S1	Scanning electron microscopy image of LMS2 (Δ*divIVA*)	Images/Figure_S1_divIVA_raw_image.tif
Figure S1	Scanning electron microscopy image of LMJR183 (i*ezrA*)	Images/Figure_S1_ezrA_raw_image.tif
Figure S1	Scanning electron microscopy image of LMSW49 (i*mreB*)	Images/Figure_S1_mreB_raw_image.tif

Supplementary Figure S2

Figure	Description	File
Figure S2	Growth curves of the strains LMSW207 (Δ*divIVA imreB*) and LMSW208 (Δ*divIVA iezrA*)	Figure_S2_raw_data.xlsx

Supplementary Figure S3

Figure	Description	File
Figure S3	Fluorescence microscopy image of macrophages infected with LMJD20 (wt, DAPI)	Images/Figure_S3_wt_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMJD20 (wt, phase contrast)	Images/Figure_S3_wt_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMJD20 (wt, TRITC)	Images/Figure_S3_wt_TRITC_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW202 (i*ezrA*, DAPI) without IPTG	Images/Figure_S3_ezrA_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW202 (i*ezrA*, phase contrast) without IPTG	Images/Figure_S3_ezrA_pc_raw_image.tif

Figure	Description	File
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW202 (*iezrA*, TRITC) without IPTG	Images/Figure_S3_ezrA_TRITC_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW203 (*imreB*, DAPI) without IPTG	Images/Figure_S3_mreB_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW203 (*imreB*, phase contrast) without IPTG	Images/Figure_S3_mreB_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW203 (*imreB*, TRITC) without IPTG	Images/Figure_S3_mreB_TRITC_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW205 (Δ*divIVA*, DAPI)	Images/Figure_S3_divIVA_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW205 (Δ*divIVA*, phase contrast)	Images/Figure_S3_divIVA_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW205 (Δ*divIVA*, TRITC)	Images/Figure_S3_divIVA_TRITC_raw_image.tif

Supplementary Figure S4

Figure	Description	File
Figure S4	Phospholipolytic activity on egg-yolk agar of EGD-e (wt)	Images/Figure_S4_egg_yolk_agar_wt_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of LMS2 (Δ*divIVA*)	Images/Figure_S4_egg_yolk_agar_divIVA_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of BUG2214 (Δ*prfA*)	Images/Figure_S4_egg_yolk_agar_prfA_raw_image.tif

Figure	Description	File
Figure S4	Phospholipolytic activity on egg-yolk agar of BUG3057 (*prfA**)	Images/Figure_S4_egg_yolk_agar_prfA-star_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_egg_yolk_agar_divIVA-prfA-star_raw_image.tif
Figure S4	Hemolytic activity on sheep blood agar of EGD-e (wt), BUG2214 (Δ*prfA*) and BUG3057 (*prfA**)	Images/Figure_S4_hemolysis_wt_prfA_prfA-star_raw_image.tif
Figure S4	Hemolytic activity on sheep blood agar of LMS2 (Δ*divIVA*) and LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_hemolysis_divIVA_divIVA-prfA-star_raw_image.tif
Figure S4	Swarming on soft agar of EGD-e (wt), BUG2214 (Δ*prfA*), BUG3057 (*prfA**), LMS2 (Δ*divIVA*) and LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_swarming_plate_raw_image.tif
Figure S4	Cellular morphology after nile red staining of EGD-e (wt)	Images/Figure_S4_nile_red_wt_raw_image.tif
Figure S4	Cellular morphology after nile red staining of LMS2 (Δ*divIVA*)	Images/Figure_S4_nile_red_divIVA_raw_image.tif
Figure S4	Cellular morphology after nile red staining of BUG2214 (Δ*prfA*)	Images/Figure_S4_nile_red_prfA_raw_image.tif
Figure S4	Cellular morphology after nile red staining of BUG3057 (*prfA**)	Images/Figure_S4_nile_red_prfA-star_raw_image.tif
Figure S4	Cellular morphology after nile red staining of LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_nile_red_divIVA-prfA-star_raw_image.tif

Metadata

To increase findability, the provided data are described with metadata. The Metadata are distributed to the relevant platforms via GitHub Actions. There is a specific metadata file for each platform; these are stored in the metadata folder:

> Metadata/

Versioning and DOI assignment are performed via Zenodo.org. The metadata prepared for import into Zenodo are stored in the zenodo.json. Documentation of the individual metadata variables can be found at https://developers.zenodo.org/representation.

> Metadata/zenodo.json

The zenodo.json includes the publication date and the date of the data status in the following format (example):

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    "dates": [
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        "end": "2023-09-11T15:00:21+02:00",
        "type": "Created",
        "description": "Date when the published data was created"
      }
    ],
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Guidelines for reuse of the data

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- https://zenodo.org/communities/robertkochinstitut
- https://github.com/robert-koch-institut
- https://gitlab.opencode.de/robert-koch-institut
- https://edoc.rki.de/

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